

Logarithmic Functions & Properties



Converting between exponential and logarithmic form

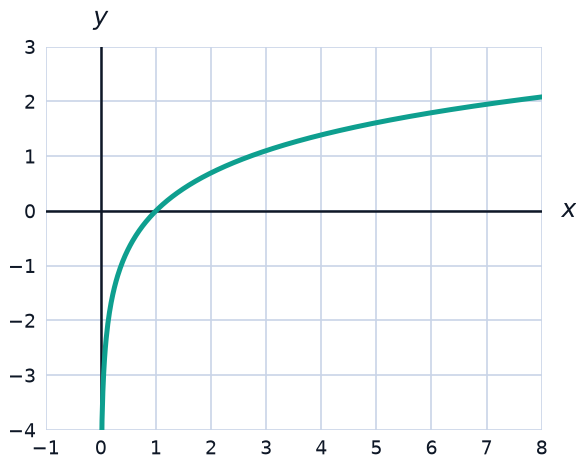
- 1 Rewrite $2^5 = 32$ in logarithmic form, and $\log_3 81 = 4$ in exponential form.
 - 2 Evaluate $\log_5 125$ and $\log_2 \frac{1}{8}$ without a calculator.
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Properties of logarithms

- 3 Use logarithm properties to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ fully.
 - 4 Combine $2\log x - \frac{1}{3}\log y + \log z$ into a single logarithm.
 - 5 Use the change of base formula to evaluate $\log_7 50$, to three decimal places.
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Visualizing logarithmic functions

- 6 The graph shows $f(x) = \ln(x)$, the natural logarithm.



- a State the domain and the vertical asymptote shown in the graph.
 - b State the x-intercept shown.
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Solving exponential equations using logarithms

- 7 Solve $5^x = 40$ for x , to three decimal places.

- 8 Solve $3(2)^{x-1} = 45$ for x , to three decimal places.
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Solving logarithmic equations

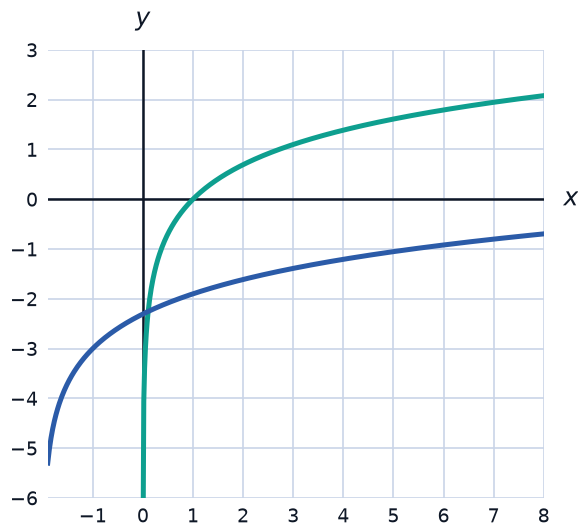
- 9 Solve $\log_2(x + 3) = 4$ for x , and check that the solution keeps the argument positive.
- 10 Solve $\log(x) + \log(x - 3) = 1$ for x , checking both solutions against the domain.
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Applications: the pH scale and decibels

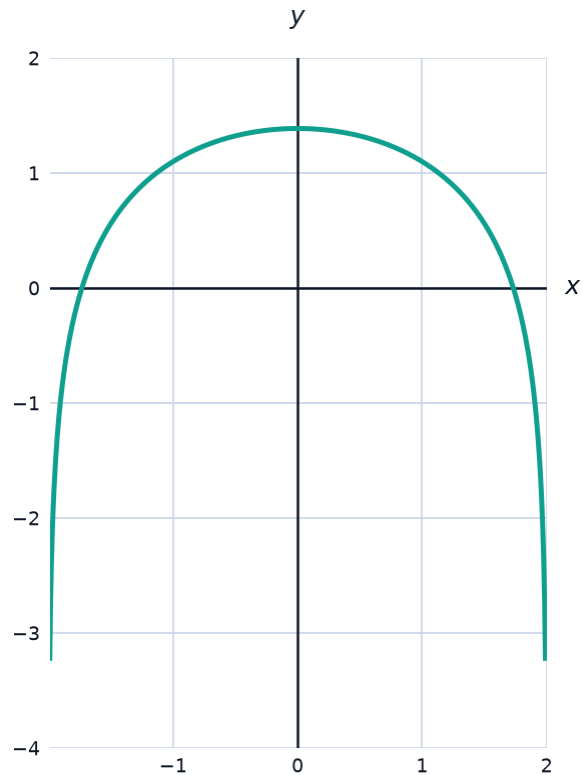
- 11 The pH of a solution is given by $\text{pH} = -\log[H^+]$, where $[H^+]$ is the hydrogen ion concentration in mol/L. Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$ mol/L, to two decimal places.
- 12 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. If one sound is 1000 times more intense than another, find the difference in their decibel levels.
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Transformations of logarithmic functions

- 13 Describe how the graph of $g(x) = \ln(x + 2) - 3$ is transformed from the parent function $f(x) = \ln(x)$, and state its new vertical asymptote and domain.



- 14 Find the domain of $h(x) = \log(4 - x^2)$.

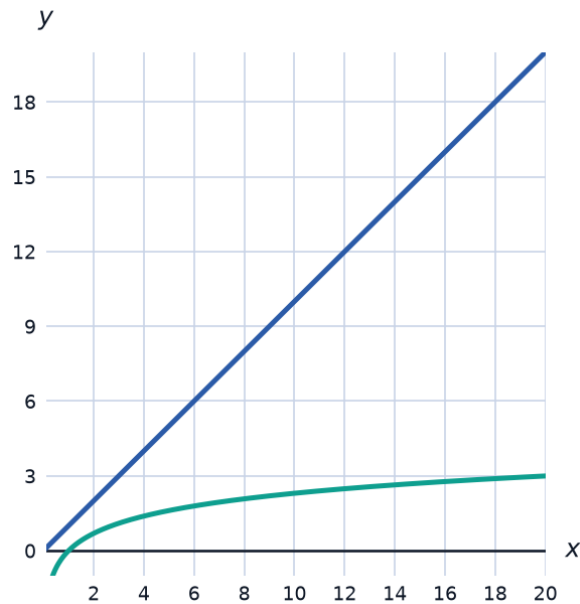


Solving equations that combine exponentials and logarithms

- 15 Solve $e^{2x} - 5e^x + 6 = 0$ by substituting $u = e^x$.
- 16 Solve $\log_2(x) + \log_2(x - 2) = 3$ for x , checking the domain.

Comparing logarithmic growth to other function families

- 17 Explain why logarithmic functions grow much more slowly than linear functions for large x , and give a numerical comparison of $\log(x)$ and x at $x = 1000$.



Modeling with logarithms: the Richter scale

- 18 Earthquake magnitude on the Richter scale is $M = \log\left(\frac{I}{I_0}\right)$, where I_0 is a reference intensity. An earthquake measures magnitude 7, while another measures magnitude 5. Find how many times more intense the magnitude-7 earthquake is.
- 19 If an earthquake's intensity is 10,000 times a reference intensity I_0 , find its Richter magnitude.

Multiple choice — easy 15

- 20 Evaluate $\log_2(2)$ without a calculator.
- a. 1
 - b. -1
 - c. 0
 - d. 2

21 Evaluate $\log_9(81)$ without a calculator.

- a. 3
- b. 1
- c. -2
- d. 2

22 Evaluate $\log_6(36)$ without a calculator.

- a. 2
- b. 1
- c. -2
- d. 3

23 Evaluate $\log_2(8)$ without a calculator.

- a. 2
- b. 4
- c. -3
- d. 3

24 Evaluate $\log_4(16)$ without a calculator.

- a. 3
- b. 2
- c. -2
- d. 1

25 Evaluate $\log_4 \left(\frac{1}{64} \right)$ without a calculator.

a. -3

b. -2

c. 3

d. -4

26 Evaluate $\log_4 (64)$ without a calculator.

a. -3

b. 3

c. 4

d. 2

27 Evaluate $\log_3 \left(\frac{1}{9} \right)$ without a calculator.

a. -1

b. 2

c. -3

d. -2

28 Evaluate $\log_6 \left(\frac{1}{216} \right)$ without a calculator.

a. 3

b. -4

c. -3

d. -2

29 Evaluate $\log_4(4)$ without a calculator.

- a. 2
- b. 0
- c. -1
- d. 1

30 Evaluate $\log_3\left(\frac{1}{3}\right)$ without a calculator.

- a. -2
- b. 0
- c. 1
- d. -1

31 Evaluate $\log_2\left(\frac{1}{2}\right)$ without a calculator.

- a. 0
- b. 1
- c. -2
- d. -1

32 Evaluate $\log_5(5)$ without a calculator.

- a. 0
- b. 2
- c. 1
- d. -1

33 Evaluate $\log_4 \left(\frac{1}{256} \right)$ without a calculator.

- a. -3
- b. -5
- c. -4
- d. 4

34 Evaluate $\log_5 (625)$ without a calculator.

- a. 4
- b. 5
- c. 3
- d. -4

Multiple choice — medium 15

35 Solve $\log_6 (x) = 3$ for x .

- a. 215
- b. 217
- c. 216
- d. 218

36 Solve $\log_4 (x - 6) = 1$ for x .

- a. -2
- b. 11
- c. 10
- d. 9

37 Solve $\log_2(x + 9) = 2$ for x .

a. 13

b. -4

c. -5

d. -6

38 Solve $\log_5(x - 5) = 3$ for x .

a. 129

b. 130

c. 120

d. 131

39 Solve $\log_3(x - 6) = 2$ for x .

a. 15

b. 3

c. 14

d. 16

40 Solve $\log_4(x + 9) = 1$ for x .

a. 13

b. -4

c. -5

d. -6

41 Solve $\log_6(x + 6) = 1$ for x .

a. -1

b. 1

c. 0

d. 12

42 Solve $\log_3(x + 4) = 2$ for x .

a. 13

b. 4

c. 5

d. 6

43 Solve $\log_4(x + 7) = 1$ for x .

a. -2

b. -3

c. -4

d. 11

44 Solve $\log_5(x + 3) = 3$ for x .

a. 123

b. 122

c. 128

d. 121

45 Solve $\log_6(x - 3) = 1$ for x .

- a. 8
- b. 9
- c. 10
- d. 3

46 Solve $\log_6(x + 2) = 2$ for x .

- a. 33
- b. 34
- c. 35
- d. 38

47 Solve $\log_2(x + 7) = 1$ for x .

- a. -5
- b. 9
- c. -4
- d. -6

48 Solve $\log_5(x - 8) = 2$ for x .

- a. 34
- b. 17
- c. 32
- d. 33

49 Solve $\log_6(x - 6) = 2$ for x .

- a. 42
- b. 43
- c. 30
- d. 41

Multiple choice — hard 15

50 Solve $4^x = 138$ for x , to three decimal places.

- a. 4.054
- b. 3.654
- c. 3.554
- d. 3.454

51 Solve $9^x = 165$ for x , to three decimal places.

- a. 2.824
- b. 2.424
- c. 2.224
- d. 2.324

52 Solve $9^x = 96$ for x , to three decimal places.

- a. 2.177
- b. 2.077
- c. 1.977
- d. 2.577

53 Solve $2^x = 109$ for x , to three decimal places.

- a. 6.868
- b. 7.268
- c. 6.668
- d. 6.768

54 Solve $6^x = 90$ for x , to three decimal places.

- a. 2.411
- b. 2.511
- c. 2.611
- d. 3.011

55 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10 times more intense than another, find the difference in their decibel levels.

- a. 30
- b. 20
- c. 10
- d. 1

56 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^6 times more intense than another, find the difference in their decibel levels.

- a. 50
- b. 70
- c. 60
- d. 6

- 57 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^3 times more intense than another, find the difference in their decibel levels.
- a. 40
 - b. 20
 - c. 3
 - d. 30
- 58 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^2 times more intense than another, find the difference in their decibel levels.
- a. 20
 - b. 10
 - c. 2
 - d. 30
- 59 Sound intensity level in decibels is $L = 10\log\left(\frac{I}{I_0}\right)$. If one sound is 10^5 times more intense than another, find the difference in their decibel levels.
- a. 5
 - b. 40
 - c. 60
 - d. 50
- 60 Solve $\log_{40}(x) + \log_{40}(x + 3) = 1$ for x , checking both solutions against the domain.
- a. -3
 - b. 6
 - c. 5
 - d. -8

- 61 Solve $\log_{21}(x) + \log_{21}(x - 4) = 1$ for x , checking both solutions against the domain.
- a. 8
 - b. -3
 - c. 7
 - d. 4
- 62 Solve $\log_{12}(x) + \log_{12}(x - 4) = 1$ for x , checking both solutions against the domain.
- a. 4
 - b. -2
 - c. 7
 - d. 6
- 63 Solve $\log_6(x) + \log_6(x - 5) = 1$ for x , checking both solutions against the domain.
- a. 6
 - b. 7
 - c. -1
 - d. 5
- 64 Solve $\log_{18}(x) + \log_{18}(x + 7) = 1$ for x , checking both solutions against the domain.
- a. 3
 - b. 2
 - c. -9
 - d. -7